DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, UNIVERSITY OF CALCUTTA

Class Test: Advanced DBMS

M. TECH. SEMESTER-I

Full Marks: 20 Time: 45 minutes

1. Justify on the following comments:
   a. Allocation schema is independent of the fragmentation schema
   b. Disjoint-ness property is not a mandatory requirement for defining the fragmentation schema of a database.

2x5=10

2. Consider the following two relations in a database. SUP(scode, sname, phone, age, city), and SP(scode, pcode, qty). It is given that all the suppliers whose records are stored in SUP belong to one of the 3 cities: Kolkata, Durgapur, and Patna.
   a. Write the fragmentation schema for SUP so that all the records for the suppliers from each of the three cities Kolkata, Durgapur, and Patna are stored in separate fragments.
   b. Define the fragmentation schema for the SP relation so that all the transactions for the suppliers from a particular city are kept in the same fragment of relation SP.
   c. Write Level 1 and Level 2 queries for the following two queries:
      Q1: The end user wants to list names and phone numbers for all such suppliers who have supplied part P7 for more than 100 units.
      Q2: Change the city for Supplier with scode = "S2205" to Patna.

2+3+(2+3)=10